

LAB3 Experiment Report
CXR multi-class classification

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1. Introduction

The objective of this lab is to develop a robust deep learning framework for the multi-class classification of Chest X-Ray (CXR) images. The task involves categorizing images into four distinct classes: **Normal**, **Bacterial Pneumonia**, **Viral Pneumonia**, and **COVID-19**.

The primary challenge is the **high inter-class similarity** between **Bacterial** and **Viral** pneumonia. Clinically and visually, these two conditions often share overlapping features (e.g., consolidation patterns), making them difficult for deep learning models to distinguish, leading to frequent misclassification. Secondly, the dataset exhibits **severe class imbalance**. The distribution is heavily skewed, with Bacterial pneumonia being the dominant class, while COVID-19 represents a tiny fraction of the data. Furthermore, there is a specific imbalance even between the two pneumonia types (Bacteria vs. Virus), which biases standard models towards the majority class.

To overcome these obstacles and maximize the macro-F1 score, I proposed a comprehensive strategy incorporating **ROI-based Preprocessing**, **Data Augmentation**, and **Semi-Supervised Learning techniques**. My approach includes:

1. **Lung Segmentation:** To eliminate background noise (e.g., bones, medical devices) and force the model to focus on lung opacities, I implemented segmentation masks to crop or mask the input images.
2. **Contrast Enhancement:** I applied **CLAHE** (Contrast Limited Adaptive Histogram Equalization) specifically on lung regions to highlight subtle texture differences between viral and bacterial infiltrates.
3. **Model Architecture & Loss Design:** I benchmarked state-of-the-art architectures (ResNet, EfficientNet, ConvNeXt) and utilized **Weighted Cross-Entropy** and **Focal Loss** to penalize the model more for misclassifying minority classes. **Label Smoothing** was also employed to prevent overfitting on noisy labels.
4. **Semi-Supervised & Post-Processing Strategy:** To leverage unlabeled information, I implemented a **Pseudo-labeling** pipeline to expand the training set. Additionally, I devised an inference weight adjustment strategy to specifically mitigate the model's tendency to misclassify Virus as Bacteria, thereby improving the recall for the harder class.

2. Implementation details

2.1 The details of model

To identify the optimal architecture for this specific chest X-ray classification task, I developed a model building framework capable of instantiating various state-of-the-art Convolutional Neural Network (CNN) families. This allowed me to experiment with different model depths and complexities to find the best trade-off between performance and computational cost.

2.1.1 Model Architectures

I focused on three major families of architectures:

- **ResNet Family (ResNet-18, 34, 50):** I implemented support for ResNet-18, 34, and 50. These models utilize **Residual Learning** to mitigate the vanishing gradient problem. ResNet-18 and 34 offer lighter baselines, while ResNet-50 provides deeper feature extraction capabilities.

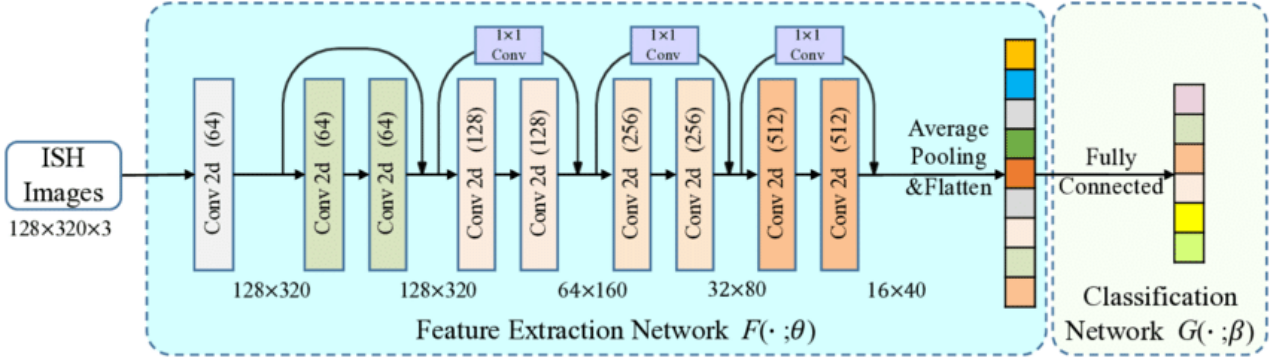


Figure 2-1 Frameworks of ResNet

- **EfficientNet Family (B0 – B2):** I experimented with the EfficientNet series from B0 to B2. These models use **Compound Scaling** to balance network width, depth, and resolution. Their **MBConv** blocks with Squeeze-and-Excitation are highly effective at capturing fine-grained texture details in medical images while maintaining parameter efficiency.

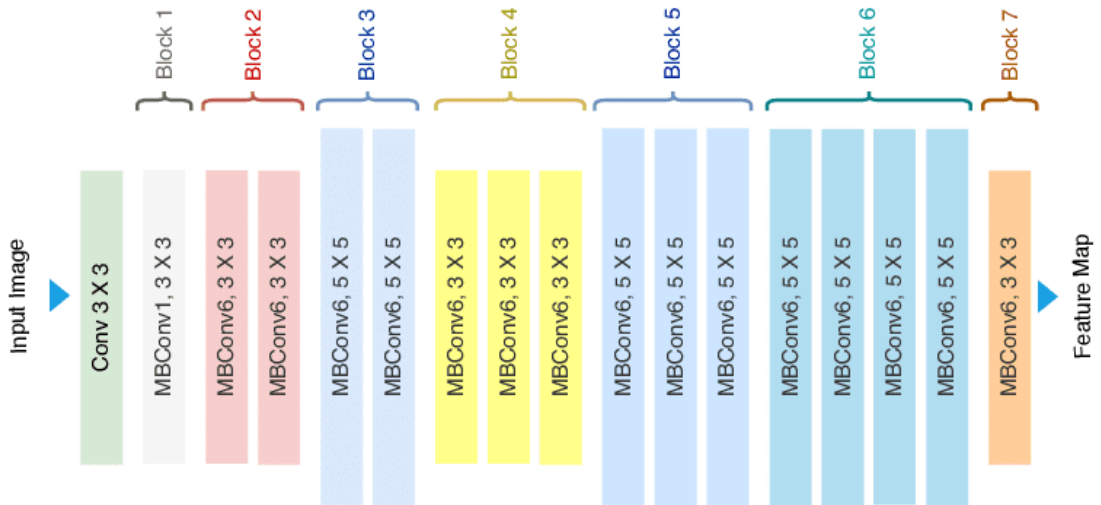


Figure 2-2 Frameworks of EfficientNet

- **ConvNeXt Family (Tiny, Small):** I included ConvNeXt (Tiny and Small variants) to benchmark modern "Transformer-like" CNNs. ConvNeXt modernizes the standard ResNet block with larger kernels (7X7) and Layer Normalization, aiming to provide the global receptive field of Vision Transformers while retaining the inductive bias of CNNs.

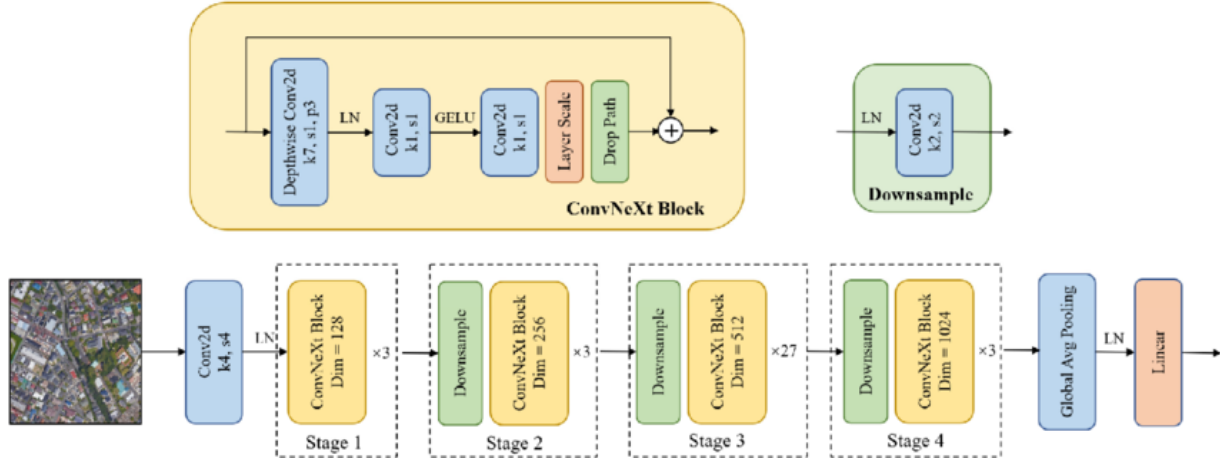


Figure 2-3 Frameworks of ConvNeXt

2.1.2 Model Settings and Customization

To adapt these ImageNet-pretrained models for our **4-class classification** task (Normal, Bacteria, Virus, COVID-19), I implemented a unified `build_model` function that handles the specific architecture modifications required for each family:

- **Transfer Learning:** All backbones are initialized with **ImageNet** weights to leverage learned low-level features.
- **Head Modification:** The final classification layer was dynamically replaced based on the architecture type:
 - ResNet: Replaced `model.fc` with `nn.Linear(in_features, 4)`.
 - EfficientNet: Replaced `model.classifier[1]` with `nn.Linear(in_features, 4)`.
 - ConvNeXt: Replaced `model.classifier[2]` with `nn.Linear(in_features, 4)`.

2.2 The details of Dataloader

To enhance the model's generalization capability and prevent overfitting—given the limited dataset size and class imbalance—I implemented a robust data augmentation pipeline using the Albumentations library. The pipeline is designed to simulate real-world clinical variations and force the model to learn invariant features.

2.2.1 Geometric Transformations (Simulating Patient Positioning)

Chest X-rays often vary due to patient posture or machine alignment. To address this, I applied:

- **Horizontal Flip** ($p=0.5$): Simulates natural anatomical mirroring.
- **Affine Transformations** ($p=0.7$): A combination of rotation, translation, and scaling was applied to simulate variations in patient distance and alignment.
 - **Rotation** ($\pm 10^\circ$): Restricted to a realistic range to maintain physiological relevance.
 - **Translation** ($\pm 10^\circ$): Simulates off-center positioning.
 - **Scaling** (90%~100%): Simulates zoom/distance variations.



Figure 2-4 Original Image vs. Horizontal Flip vs. Affine (Rotate+Translate+Scale)

2.2.2 Pixel-level Enhancement (Medical Feature Extraction)

Distinguishing between Bacterial (focal consolidation) and Viral (diffuse interstitial patterns) pneumonia requires the model to recognize subtle texture differences. I applied one of the following transformations with a probability of 0.8:

- **CLAHE (Contrast Limited Adaptive Histogram Equalization)**: With clip limit=2.0 and tile_grid_size= (8, 8). This significantly enhances local contrast, making subtle lung opacities and infiltrates more visible against the ribs and soft tissue.
- **Random Gamma & Sharpening**: Simulates different exposure settings and detector sensitivities found in various X-ray machines.



Figure 2-5 CLAHE vs. Random Gamma vs Sharpen

2.2.3 Regularization via Occlusion (Coarse Dropout)

To prevent the model from overfitting to specific local artifacts (e.g., medical tubes, tags) or relying solely on a single focal area, I implemented **coarse Dropout** (a technique similar to Cutout).

- **Settings:** Randomly generates 4 to 8 rectangular black holes on the image.
- **Impact:** This forces the network to utilize global context and distributed features across the entire lung field to make predictions, which is crucial for identifying diffuse viral pneumonia.

2.2.4 Preprocessing and Normalization

- **Resizing:** Images were resized to fixed dimensions (e.g., 224×224) to match the backbone architecture.
- **Normalization:** Pixel values were normalized using the standard ImageNet mean $([0.485, 0.456, 0.406])$ and standard deviation $([0.229, 0.224, 0.225])$ to align with the pre-trained weights.

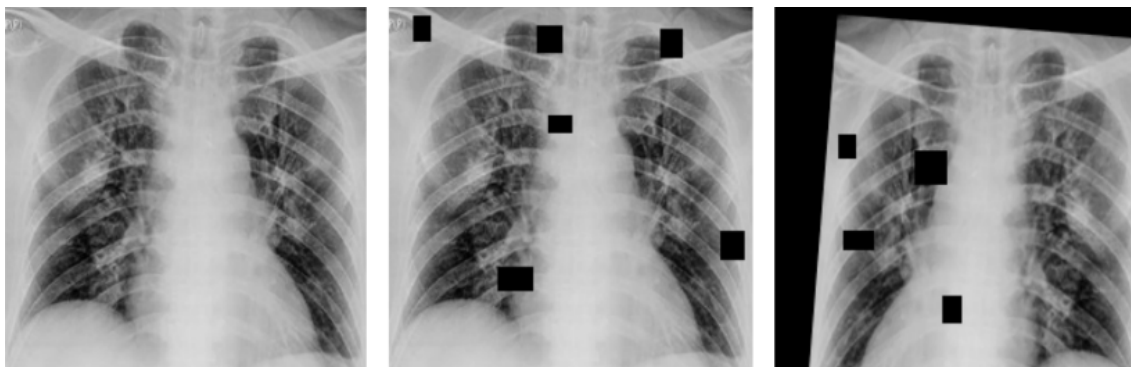


Figure 2-6 Original Image vs. Coarse Dropout vs. Combined Augmentations

3. Strategy design

3.1 Data preprocessing

In medical image analysis, raw X-ray images often contain significant "noise" in the form of irrelevant anatomical structures (e.g., clavicles, spine, shoulders) and medical artifacts (e.g., ECG leads, tubes). These distractions can prevent the model from focusing on the lung parenchyma where pneumonia indicators (consolidations and infiltrates) are located. To address this, I designed a preprocessing pipeline focusing on **Region of Interest (ROI) Extraction** and **Targeted Contrast Enhancement**.

3.1.1 ROI Extraction via Lung Segmentation

I hypothesized that removing background noise would improve classification performance, particularly for the Virus class, which relies on identifying subtle interstitial patterns. I employed a publicly available pre-trained U-Net segmentation model (<https://github.com/IliaOvcharenko/lung-segmentation>) to generate probabilistic lung mask with pixel values ranging from [0,1], where higher values indicate stronger confidence that a pixel belongs to lung tissue. Figure 3-1 illustrates an example of the segmentation output, including the original CXR, the generated lung mask, and an overlay visualization.

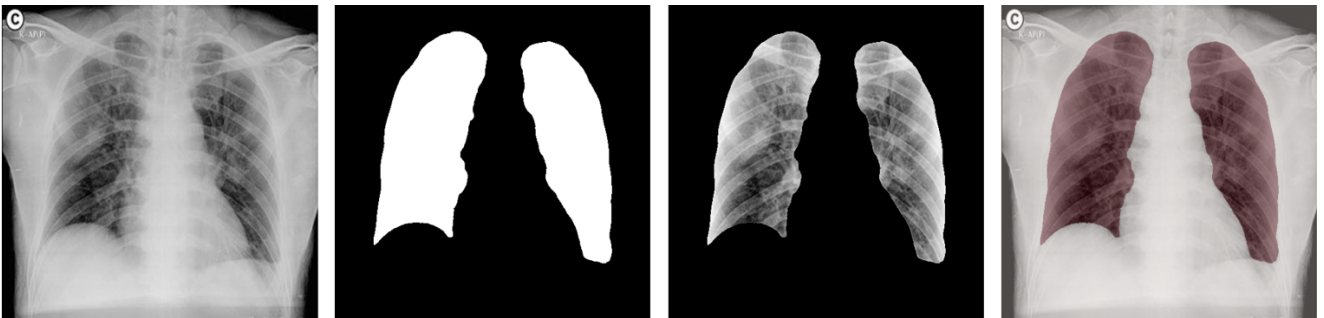


Figure 3-1 Original Image vs. Segmentation Mask vs. Segmented Lung vs. Overlay

To determine the most effective way to utilize the segmentation output for CXR classification, two different strategies were tested:

(1) Segmentation Mask $\times$ Original Image

In this approach, the lung mask is element-wise multiplied with the original image, forcing non-lung areas to zero (black).

Although this effectively removes background structures, visual inspection revealed that discontinuous or incomplete masks could result in **fragmenting the lung region**, especially when boundaries are unclear or the segmentation model deviates from the actual anatomy. This fragmentation disrupts the anatomical continuity, which may negatively affect the model's ability to learn global features. Figure 3-2 visualizes the result of a failed segmentation mask.



Figure 3-2 Original Image vs. Segmentation Mask vs. Overlay

(2) Minimum Bounding Box Cropping

To avoid the fragmented-lung issue, this experiment applies minimum bounding box cropping. Complete method as following: calculated the bounding box coordinates covering the segmented lung area and cropped the original image accordingly. As shown in Figure 3.3, this approach preserves the overall anatomical continuity of the lung region while effectively suppressing irrelevant background (e.g., shoulders, abdomen). This was selected as the final ROI strategy.

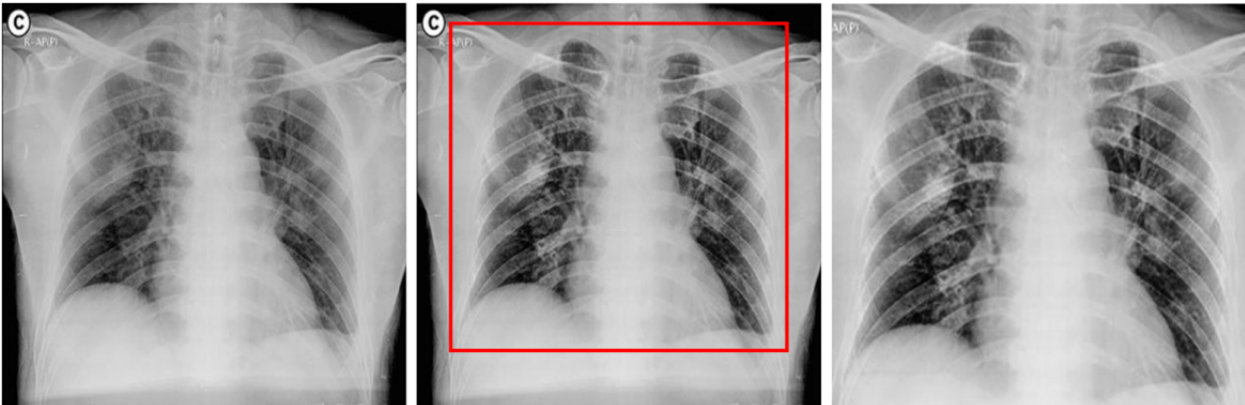


Figure 3-3 Original Image vs. Minimum Bounding Box vs. Cropping Result

3.1.2 Image Enhancement (CLAHE)

To highlight subtle pneumonia-related patterns (e.g., infiltrates, ground-glass opacity), I applied **Contrast Limited Adaptive Histogram Equalization (CLAHE)**. Two distinct application strategies were compared:

(1) Full-Image CLAHE

CLAHE is applied to the entire CXR. While overall contrast is improved, non-lung regions such as bones, medical devices, and background noise are also enhanced. These enhanced artifacts act as distractions for the classification model.

(2) Lung-Only CLAHE

By applying CLAHE specifically to the cropped lung ROI, the histogram equalization is calculated based solely on lung tissue pixel statistics. This maximizes the contrast in the relevant soft tissue regions without amplifying background noise.



Figure 3-4 Original Image vs. Full Image After CLAHE vs. Cropped Image After CLAHE

3.2 What makes training strategy special?

To significantly outperform standard supervised learning baselines, I engineered a specialized training framework designed to handle the dataset's specific pathologies: extreme class imbalance (COVID-19 < 1%) and high inter-class confusion (Bacteria vs. Virus).

3.2.1 Loss Function Design

(1) Baseline Analysis & Problem Identification

In the preliminary phase, I established a baseline using EfficientNet-B2 trained with standard Cross-Entropy (CE) loss, which treats all samples equally. This baseline achieved a Macro F1-score of 0.8924. While the model performed exceptionally well on Normal and COVID-19 classes, the performance on Bacteria and Virus was significantly lower. The confusion matrix indicated a high rate of mutual misclassification between these two pneumonia types, identifying this overlap as the primary bottleneck.

(2) The Strategy for Class Weighting

To address the imbalance, I initially considered standard Inverse-Frequency Weighting. However, due to the extreme scarcity of COVID-19 samples (<1%), the mathematical formula assigned an excessive weight to this class (approximately 26x). However, In the preliminary phase showed that even without specific weighting, the baseline model could classify COVID-19 with near-perfect accuracy (F1 ~1.0). Therefore, assigning such a massive weight to COVID-19 was unnecessary and potentially destabilizing.

Instead, I adopted a Weighting Strategy. Since the Virus class consistently yielded the lowest F1-score, I prioritized it by manually assigning it the highest penalty. For example, in my experiments, misclassifying a virus sample was set to incur a penalty 4 times higher than the majority classes, forcing the model to focus on these hard examples.

(3) Loss Functions Evaluated

Based on this logic, I experimented with three specific loss functions to tackle the confusion and imbalance:

A. Weighted Cross-Entropy (WCE)

I applied the weights mentioned above (e.g., Virus=4.0, Bacteria=1.0) to directly penalize the model for missing viral cases.

B. Focal Loss

To automatically address the "hard" examples (Virus/Bacteria confusion) without manually tuning every weight, I tested Focal Loss. It adds a modulating factor $(1 - p_t)^\gamma$ to standard CE, which down-weights easy examples (Normal/COVID) and focuses training on the difficult, misclassified ones.

C. Label Smoothing Cross-Entropy

Given the visual ambiguity between bacterial and viral pneumonia, "hard" labels (0 or 1) might force the model to be overconfident. Label Smoothing introduces a tolerance $\varepsilon = 0.1$, encouraging the model to learn smoother decision boundaries and improving generalization on fuzzy samples.

3.2.2 Semi-Supervised Learning: Pseudo-Labeling

Given the limited training data for specific classes and the availability of the unlabeled test set, I employed a Pseudo-Labeling pipeline to perform domain adaptation as illustrated in Figure 3-5.

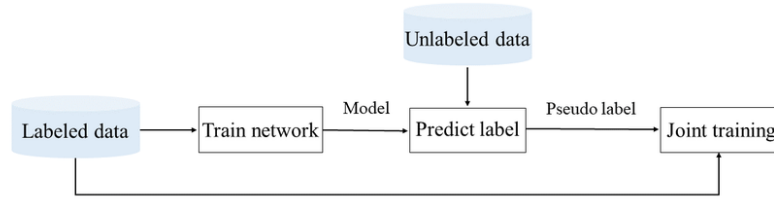


Figure 3-5 The Pseudo-Labeling Workflow

(1) Teacher Training: Teacher Training & Selection

First, a robust baseline model is trained on the labeled training set. The selection of this "Teacher" model was strictly determined by the validation **Macro F1-Score**.

(2) Inference & Filtering

The Teacher model predicts labels for the unlabeled test set. A strict **Confidence Threshold** is applied. Since the approximate data distribution of the test set was known, this information was used to verify the validity and quality of the filtered pseudo-labels, ensuring no extreme bias was introduced.

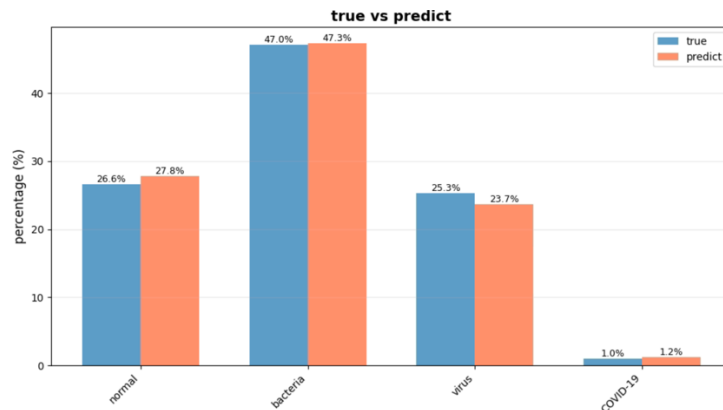


Figure 3-6 Compare distribution between Test datasets and predicted label

(3) Dataset Expansion

Only test images where the model is >90% confident are treated as "ground truth" and merged into the training set.

(4) Student Retraining

Finally, a new "Student" model is trained from scratch on this expanded dataset (Original Training Set + Pseudo-labeled Samples). This allows the model to learn a decision boundary that better generalizes to the test domain.

3.3 Experimental Settings

Category	Parameter	Value/Setting
Optimization	Optimizer	AdamW
	Learning Rate	3e-4
	Weight Decay	1e-2
Scheduling	Scheduler	CosineAnnealingLR
Training Loop	Epochs	50
	Batch Size	32
	Input Size	384x384
	Early stop	15
Loss Function	Cross Entropy	
	WCE	cw= [1.0, 1.0, 4.0, 1.5]
	Focal Loss	gamma = 2.0, cw= [1.0, 1.0, 4.0, 1.5]
	Label smooth CE	epsilon = 0.1

4. Discussion

4.1 Baseline Experiment Analysis: The "Virus" Bottleneck

To determine the optimal backbone and preprocessing strategy, I benchmarked eight different architectures under two conditions: **Full Image + CLAHE** versus **Cropped Image + CLAHE**. The detailed class-wise performance metrics are summarized in Table 4-1.

Table 4-1 Impact of Preprocessing on Bacteria and Virus Classification

Preprocess method	Full image + CLAHE			Cropped image + CLAHE		
F1 score	Macro	Bacteria	Virus	Macro	Bacteria	Virus
Resnet18	0.8647	0.8675	0.7273	0.8741	0.8720	0.7198
Resnet34	0.8484	0.8370	0.7003	0.8632	0.8567	0.7374
Resnet50	0.8698	0.8375	0.7028	0.8687	0.8550	0.7011
Efficientnet-B0	0.8907	0.8759	0.7347	0.8868	0.8690	0.7356
Efficientnet-B1	0.8988	0.8746	0.7655	0.8915	0.8653	0.7624
Efficientnet-B2	0.8777	0.8520	0.7111	0.8924	0.8798	0.7640
ConvneXt-tiny	0.8637	0.8401	0.6857	0.8813	0.8625	0.7302
ConvneXt-small	0.8721	0.8580	0.7500	0.8769	0.8571	0.7216

4.1.1 The Challenge of "Virus" Classification

Aside from the **COVID-19** and **Normal** classes, which consistently achieved stable F1-scores above **0.9** across all models, the most consistent finding across all 16 experiments is the comparatively lower performance of **Bacteria** and **Virus**, and the significant performance gap between them.

- Bacteria F1-Score: Ranging from 0.83 to 0.88.
- Virus F1-Score: Ranging from 0.68 to 0.76.

This quantitative gap confirms that **Viral Pneumonia is the hardest class to learn**. Also in confusion matrices suggests that the model frequently misclassifies Virus as Bacteria. This is likely because Bacterial pneumonia presents as distinct focal consolidations, whereas Viral pneumonia manifests as diffuse interstitial patterns. These diffuse patterns are subtle and can be easily confused with "Normal" tissue or "Bacterial" consolidations depending on severity, making the decision boundary extremely noisy.

4.1.2 The "Cropping" Trade-off: Noise Removal vs. Global Context

While **Cropping** generally improved the overall **Macro F1** (by improving Normal and COVID-19 detection via noise removal), its impact on the critical **Virus** class was heterogeneous:

- **Models that Benefited (e.g., EfficientNet-B2, ConvNeXt-Tiny):** For these models, Cropping significantly improved Virus F1 (e.g., +5.29% for EfficientNet-B2). This suggests that

removing background noise (clavicles, medical devices) helped these architectures focus on the subtle texture of viral infiltrates within the lungs.

- **Models that Degraded (e.g., ConvNeXt-Small, ResNet-18):** Conversely, for models like ConvNeXt-Small, Cropping hurt Virus performance (dropping from 0.7500 to 0.7216).
- **Analysis:** Since viral pneumonia is diffuse, it relies on global context. Aggressive cropping might alter the aspect ratio or cut off peripheral cues that these specific models relied on to distinguish the overall lung pattern from focal bacterial cases.

4.1.3 Selection of the Best Baseline

Based on this ablation study, **EfficientNet-B2 with Cropping** (Macro: 0.8924, Virus: 0.7640) and **EfficientNet-B1 with Full Image** (Macro: 0.8988, Virus: 0.7655) emerged as the top performers. This indicates that EfficientNet's MBConv blocks are inherently better at extracting the fine-grained features required for Virus detection compared to ResNet or ConvNeXt on this dataset.

4.2 Loss Function: WCE vs. Focal Loss vs. Label-Smoothing CE

To systematically understand how different loss formulations influence the classification of Bacterial and Viral pneumonia—the two most problematic classes—I conducted a controlled comparison among three loss functions: Weighted Cross-Entropy (WCE), Focal Loss, and Label-Smoothing Cross-Entropy.

Across all experiments, the Virus class remained the most challenging due to its diffuse and low-contrast interstitial patterns, which often resemble either mild bacterial consolidations or almost-normal lung textures. This inherent ambiguity makes the optimization highly sensitive to the loss design.

(1) Weighted Cross-Entropy (WCE)

WCE directly adjusts the gradient contribution from each class. By assigning a significantly larger weight to the Virus class (e.g., $4\times$ compared to majority classes), the model becomes more sensitive to viral misclassification. In practice, WCE produced the highest Virus F1-score (0.7850) among the three losses, confirming that explicit emphasis on this minority/hard class helps the model learn a cleaner decision boundary between Bacterial and Viral pneumonia.

(2) Focal Loss

Focal Loss was tested to automatically down-weight easy examples (Normal and COVID-19) and focus training on the ambiguous Bacteria/Virus samples. While Focal Loss improved Bacteria F1 (0.8953), it yielded a slightly lower Virus F1 (0.7697), suggesting that the dynamic down-weighting mechanism may not be as effective when the confusion is structural (Bacteria–Virus similarity) rather than simply due to class imbalance.

(3) Label-Smoothing CE

Label smoothing was initially expected to help by preventing overconfidence in visually ambiguous cases. However, it produced the lowest Virus F1 (0.7738) and notably weakened Bacteria F1 (0.8806). This indicates that smoothing the labels introduces excessive tolerance between Bacteria and Virus, further blurring an already difficult class boundary.

Among the three losses, **Weighted Cross-Entropy demonstrated the best balance**, particularly for improving Virus detection, which was the main bottleneck of the task. For this reason, WCE was selected as the teacher model used to generate pseudo-labels in the next stage.

4.3 Impact of Pseudo-Labeling

After identifying the strongest loss functions for each pneumonia type—WCE for Virus and Focal Loss for Bacteria—I employed these specialized models to generate high-confidence pseudo-labels for the unlabeled test data. Only predictions with **confidence > 90%** were accepted to ensure label quality and prevent error propagation.

4.3.1 Pseudo-Label Generation Strategy

- The **WCE-trained model** was used to generate pseudo-labels for **Virus**, taking advantage of its superior performance on this class.
- The **Focal Loss model** was used to extract high-confidence pseudo-labels for **Bacteria**, leveraging its strong Bacteria F1-score.
- Additionally, a **Label-Smoothing model** with the strongest Normal-class performance was included to enrich the **Normal** pseudo-labels.

The final pseudo-labeled set was merged with the original training data, resulting in an expanded dataset more aligned with the test distribution.

4.3.2 Effect on Model Performance

After retraining the models with the augmented dataset, the following behaviors were observed, **WCE** retraining maintained stable performance (Macro 0.9003), with minor fluctuations across classes. **Focal Loss** showed a small performance drop, likely due to noise introduced in ambiguous B/V cases. **Label smoothing** suffered the largest degradation, indicating that this loss is less robust to pseudo-label noise.

Interestingly, none of the loss functions surpassed the original baseline after pseudo-labeling. This suggests that although pseudo-labeling increased the training set size, the *quality of pseudo-labels* in ambiguous regions (especially Virus vs. Bacteria) remains a limiting factor. The structural overlap between these two classes means that even high-confidence predictions may contain subtle noise.

4.4 Future Directions and Potential Optimizations

Although the final model achieved strong macro-F1 performance, several promising optimization directions remain:

(1) Two-Stage Classification (Specialist Models)

Introduce a second-stage binary classifier dedicated to Bacteria vs. Virus. This could refine decisions in the ambiguous region where most misclassifications occur.

(2) Advanced Loss Functions (LDAM, Class-Balanced Loss)

These losses explicitly account for long-tailed distributions and margin adjustment, potentially improving discrimination of minority or hard-to-separate classes.

(3) Model Ensembling / Test-Time Augmentation

Combining predictions from multiple models or augmentations may stabilize the decision boundary and reduce false positives between Bacteria and Virus.

Overall, this project demonstrates that for medical imaging tasks involving visually overlapping disease entities, optimization must go beyond increasing dataset size. Instead, techniques such as targeted loss design, hierarchical inference strategies are likely to yield the greatest performance gains.

5. GitHub

The complete implementation is available on GitHub:

https://github.com/jhihyawang/NYCU_114Autumn_AIMI/tree/main/Lab3